



Deepfake-News Shield: Detecting False Articles and Manipulated Photos

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1. INTRODUCTION

The rise of social media and online platforms has made the spread of fake news a critical challenge. Malicious news, often crafted using text, images, and videos, can mislead the public, manipulate opinion, and cause social and political unrest. Traditional detection methods relying only on text are insufficient as fake news is now highly multimodal. This project leverages machine learning and deep learning techniques to detect fake news using textual content along with associated images, improving detection accuracy and overall system reliability significantly.

2. PROBLEM STATEMENT

Existing fake news detection systems primarily focus on text-only analysis, ignoring visual information which plays a vital role in verifying authenticity. The challenge is to design an efficient multimodal model that can jointly analyze text and image features to accurately classify news as real or fake.

3. OBJECTIVES

- To collect and preprocess a multimodal dataset of real and fake news.
- To extract meaningful features from both text and images.
- To design and implement machine learning and deep learning models for fake news detection.
- To perform comparative analysis of multiple models.
- To deploy the best performing model in a web application.

4. DATASET

Source	Kaggle – IFND Dataset
Classes	Real and Fake
Total Samples	56,714
Real Samples	37,800
Fake Samples	18,914
Split (Train/Val/Test)	80% / 10% / 10%

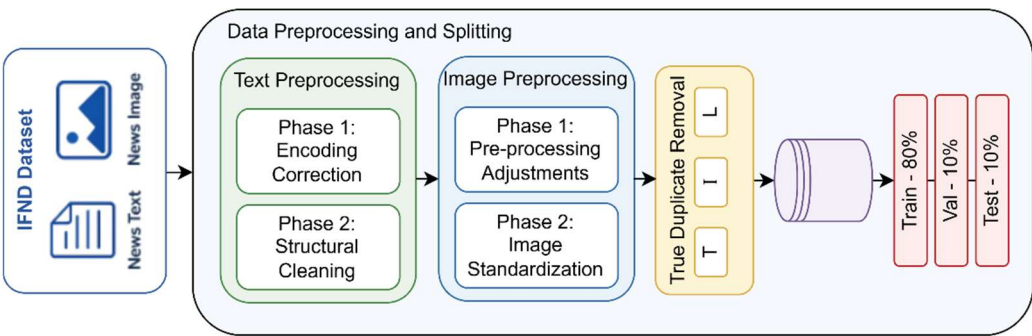


5. METHODOLOGY



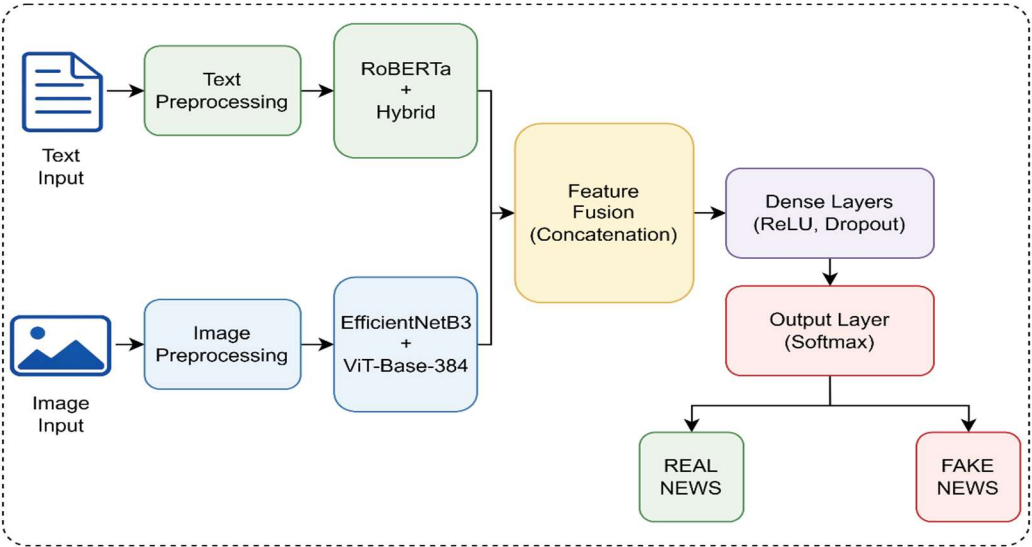
- Text and image preprocessing for data normalization.
- Feature extraction and model training for fake news classification.

6. DATA PREPROCESSING



Cleaning, standardizing, and splitting multimodal data for training.

7. SYSTEM ARCHITECTURE

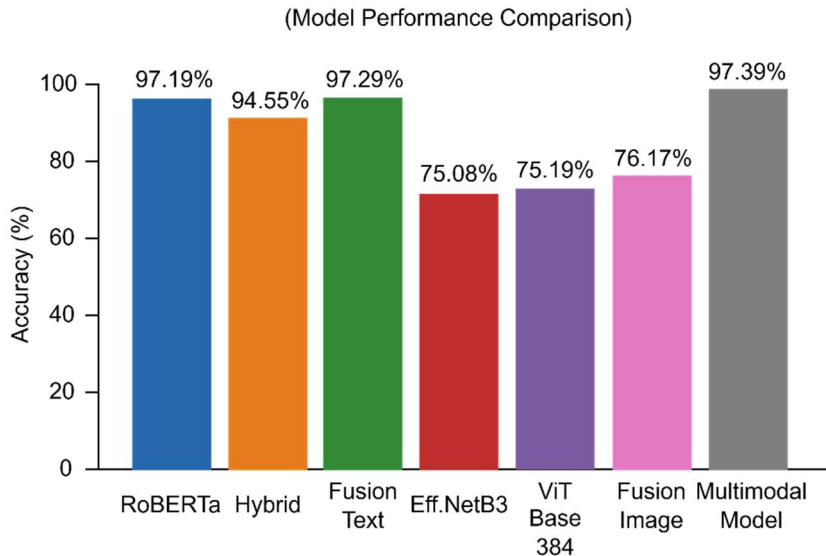


Fusion of text and image features for fake news detection.

8. MODELS USED

- Text Models: RoBERTa, Hybrid(BiLSTM+Attn+GRU).
- Image Models: EfficientNetB3, ViT-Base-384.

9. RESULTS AND DISCUSSION



The proposed Multimodal (V7) model achieved the highest accuracy of 97.39%, Macro-F1 Score of 96.98%, and AUC-Score of 98.73%, outperforming all individual text and image models.

10. FUTURE SCOPE

- Extend the model to multilingual and cross-country datasets.
- Use advanced multimodal models like CLIP and BLIP.
- Add cross-modal attention for better text-image understanding.
- Expand the framework for video and audio misinformation detection.

11. REFERENCES

- [1] D. K. Sharma and S. Garg, "IFND: a benchmark dataset for fake news detection," Complex & Intelligent Systems, 2023.
- [2] W. Y. Wang, "Liar, Liar Pants on Fire: A New Benchmark Dataset for Fake News Detection," ACL Workshop, 2017.
- [3] S. Shah and S. Patel, "A Comprehensive Survey of Fake News Detection Techniques," Journal of Computer Science, vol. 21, no. 4, pp. 982–990, 2025.
- [4] W. Y. Wang, "Liar, Liar Pants on Fire: A New Benchmark Dataset for Fake News Detection," Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics, pp. 423–428, 2017.
- [5] JGraph Ltd., draw.io Diagramming Application, 2025. [Online].